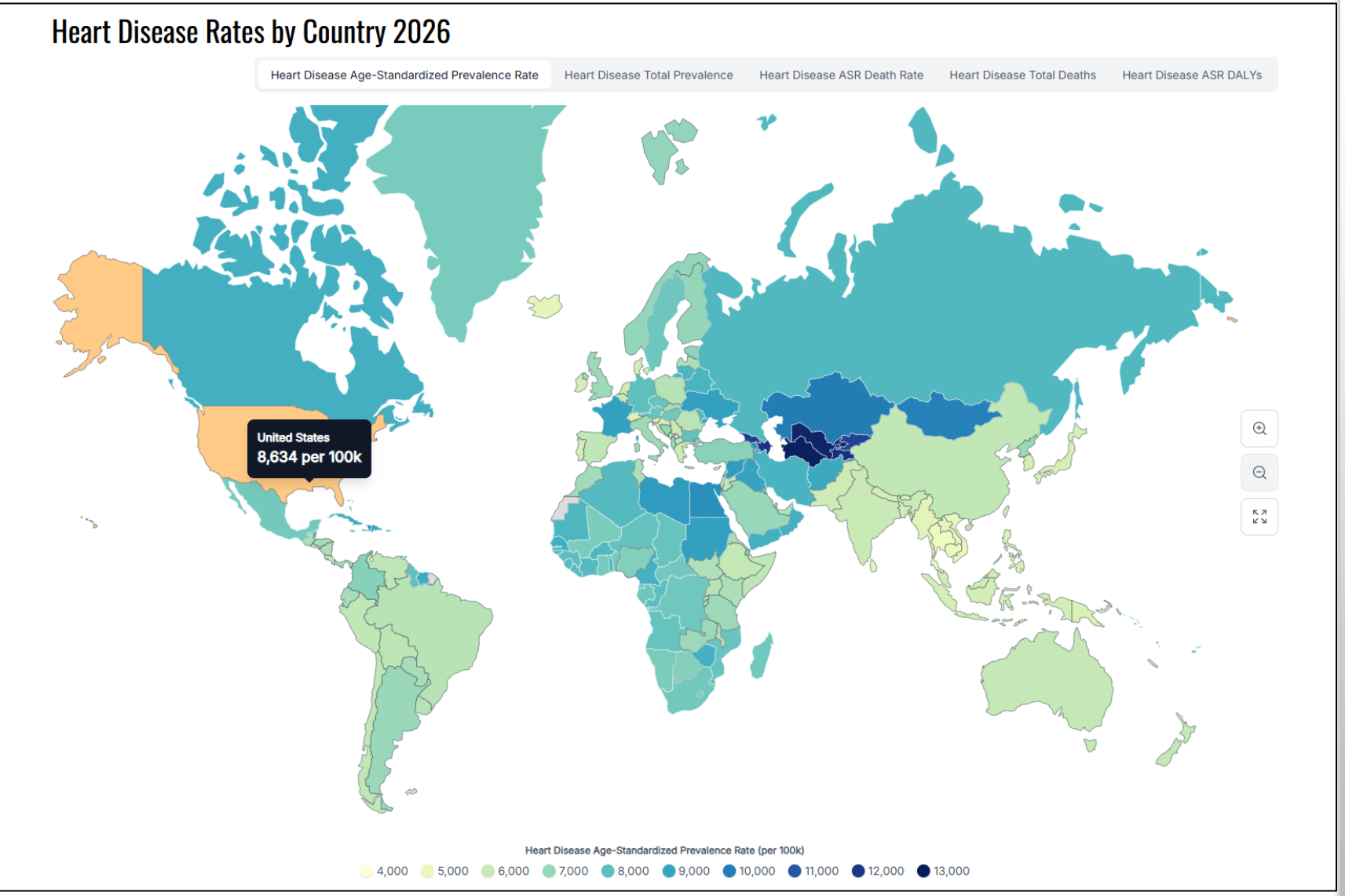
HEART DISEASE FEATURE ANALYSIS

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- Overview
- Dataset
- EDA and Visualization 1
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OVERVIEW

Heart disease is one of the leading causes of death globally, yet many people are unaware of their risks until severe symptoms appear. The prevalence of heart disease in the United States has reached 8,634 per 100,000 population. My target audience includes health-conscious adults, patients with risk factors, and healthcare students.

Main analytical question: Which key variables best predict heart disease.



DATASET

My dataset contains approximately 10,000 records. The observation unit is a single patient. The dataset covers three dimensions: demographics (Age, Gender), clinical metrics (BMI, Blood Pressure, Cholesterol, Triglyceride, Homocysteine), and lifestyle habits (Smoking, Alcohol). The target variable is 'Heart Disease Status' (Yes/No).

Age	Gender	Blood Pres	Cholestero	Exercise Ha	Smoking	Family Hea	Diabetes	BMI	High Bloo	Low HDL C	High LDL C	Alcohol Co	Stress Leve	Sleep Hou	Sugar Con	Triglyceride	Fasting Blo	CRP Level	Homocyste	Heart Disease Status
56	Male	153	155	High	Yes	Yes	No	24.99159	Yes	Yes	No	High	Medium	7.633228	Medium	342		12.96925	12.38725	No
69	Female	146	286	High	No	Yes	Yes	25.2218	No	Yes	No	Medium	High	8.744034	Medium	133	157	9.355389	19.29888	No
46	Male	126	216	Low	No	No	No	29.85545	No	Yes	Yes	Low	Low	4.44044	Low	393	92	12.70987	11.23093	No
32	Female	122	293	High	Yes	Yes	No	24.13048	Yes	No	Yes	Low	High	5.249405	High	293	94	12.50905	5.961958	No
60	Male	166	242	Low	Yes	Yes	Yes	20.48629	Yes	No	No	Low	High	7.030971	High	263	154	10.38126	8.153887	No
25	Male	152	257	Low	Yes	No	No	28.14468	No	No	No	Low	Medium	5.504876	Low	126	91	4.297575	10.81598	No
78	Female	121	175	High	Yes	Yes	Yes	18.04233	No	Yes	No	Medium	Medium	9.240911	Medium	107	85	11.58298	19.65946	No
38	Female	161	187	Low	Yes	Yes	Yes	34.73668	No	No	No	Low	Medium	7.841008	High	228	111	4.929381	17.1466	No
56	Female	135	291	Low	No	Yes	Yes	34.49311	Yes		Yes	High	Low	6.941403	High	317	103	5.119015	6.051129	No
75	Male	144	252	Low	Yes	Yes	No	30.14215	No	No	Yes	Low	Medium	4.002662	High	199	96	10.0057	7.604357	No
36	Female	179	191	High	No	Yes	No	34.44762	No	Yes	No	Medium	High	9.151889	High	231		13.58335	13.78372	No
40	Female	134	296	High	No	Yes	Yes	31.73962	No	No	No	None	Medium	6.765162	Medium	109	139	6.694785	19.44065	No
28	Female	143	215	High	Yes	No	Yes	33.34402	No	No	Yes	Medium	Low	7.188455	Low	196	142	8.426725	11.14641	No
28	Female	134	191	High	Yes	No	No	19.42243	Yes	Yes	No	Low	Low	8.189629	Low	178	102	3.380839	17.47815	No
41	Female	150	187	High	No	Yes	No	37.38784	Yes	No	No	Medium	Low	9.624732	Low	343	124	10.05179	9.897205	No
70	Male	133	290	Medium	Yes	Yes	Yes	32.16649	No	No	Yes	Medium	Low	4.94926	Medium	397	155	4.620012	7.919947	No
53	Female	173	255	Medium	No	Yes	No	31.43388	Yes	No	No	Medium	Medium	4.427105	Medium	212	160	13.30907	9.784993	No
57	Female	133	245	Low	No	Yes	Yes	26.5193	No	Yes	No	Medium	High	4.731453	High	384	132	0.028436	14.30662	No
41	Male	125	219	Medium	Yes	Yes	No	18.61898	No	Yes	Yes	High	High	8.444345	Medium	346	106	7.176291	13.86899	No
20	Female	133	187	High	Yes	No	Yes	37.83216	Yes	No	Yes	None	High	4.775667	Low	290	144	12.47282	11.70383	No
39	Female	136	246	High	Yes	Yes	No	39.68463	No	No	Yes	Medium	High	7.005886	Low	380	124	4.407841	15.3251	No
70	Female	137	257	High	No	No	No	35.77131	Yes	Yes	Yes	Low	Low	8.23897	Medium	312	135	7.27561	8.566388	No
19	Female	139	268	Low	No	Yes	No	38.13086	Yes	Yes	Yes	High	High	8.276571	High	178	80	1.586211	6.211537	No
41	Female	170	300	Low	No	No	No	20.66106	No	Yes	Yes	None	Low	6.438116	Low	317	116	1.739325	14.22584	No
61	Male	136	223	Low	Yes	No	Yes	20.83085	Yes	Yes	Yes	High	High	8.191687	Medium	102	157	9.559221	7.093717	No
47	Male	135	206	Low	No	No	No	37.35134		Yes	No	Medium	High	9.700067	Low	257	122	12.12184	12.67849	No
55	Male	159	259	Low	Yes	No	Yes	19.26513	Yes	Yes	Yes	Low	High	4.167873	High	180	146	0.907313	10.10303	No
19	Female	158	228	Low	Yes	Yes	Yes	20.05048	Yes	No	Yes	Low	High	8.028281	Low	233	126	3.19783	5.486386	No
77	Male	171	300	Medium	No	Yes	Yes	35.97064	No	Yes	Yes	High	High	7.135709	Low	176	101	6.460138	11.9334	No
38	Female	151	185	High	Yes	Yes	No	30.25344	No	Yes	Yes	Low	Low	5.275531	Medium	396	95	14.37539	18.14965	No
50	Female	179	206	Medium	No	Yes	Yes	26.618	Yes	No	Yes	Low	Medium	5.846273	Low	179	151	1.437823	18.22809	No
29	Female	163	278	Low	Yes	Yes	Yes	30.17587	Yes	No	Yes	None	Medium	9.022903	Low	292	90	11.45377	15.90084	No
75	Male	128	152	High	Yes	No	Yes	37.33245	Yes	Yes	No	Low	Low	7.521389	Medium	300	104	9.172698		No
39	Male	146	163	High	No	Yes	Yes	34.65691	Yes	No	Yes	Low	High	4.087182	Low	220	96	6.305143	10.18831	No
78	Male	165	183	High	Yes	No	Yes	33.21536	Yes	Yes	No	High	Low	9.792168	Medium	341	129	4.397678	5.460017	No
61	Female	171	219	Medium	No	Yes	No	39.49148	Yes	No	Yes	High	Low	7.997106	High	277	94	5.622685	15.82564	No

DATASET

During data preparation, I made two key decisions. First, since the missing rate was under 0.5%(except Alcohol Consumption), I used Complete Case Analysis to drop a few missing rows, which is highly defensible.

Dataset Overview

Total Valid Records	Heart Disease Cases	No Heart Disease	Heart Disease Rate
9,762	1,954	7,808	20.02%

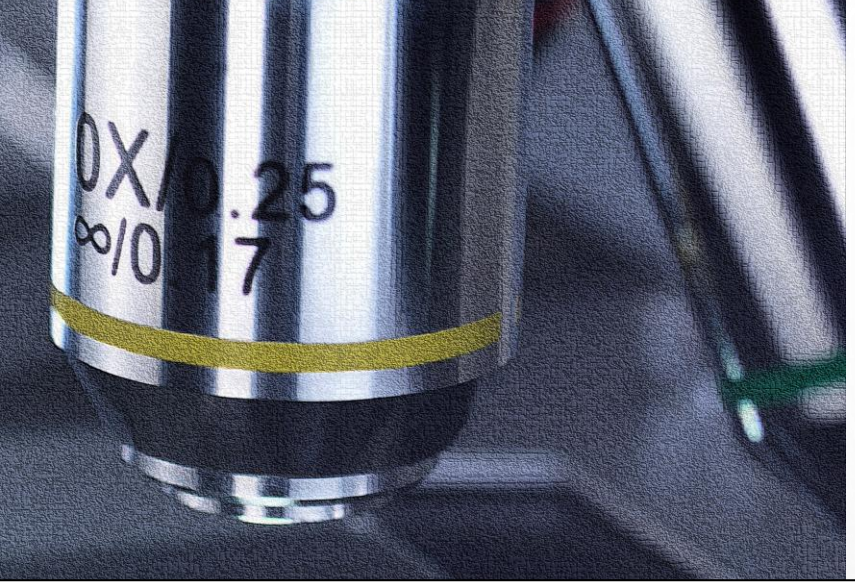
I also added a switch in Streamlit to control whether to include the alcohol consumption variable.

Dataset Overview

Total Valid Records	Heart Disease Cases	No Heart Disease	Heart Disease Rate
7,236	1,465	5,771	20.25%

Second, I found a severe class imbalance: 80% are healthy and only 20% have heart disease. A naive model guessing "No" achieves 80% accuracy but terrible false negatives, which makes me predict hard.

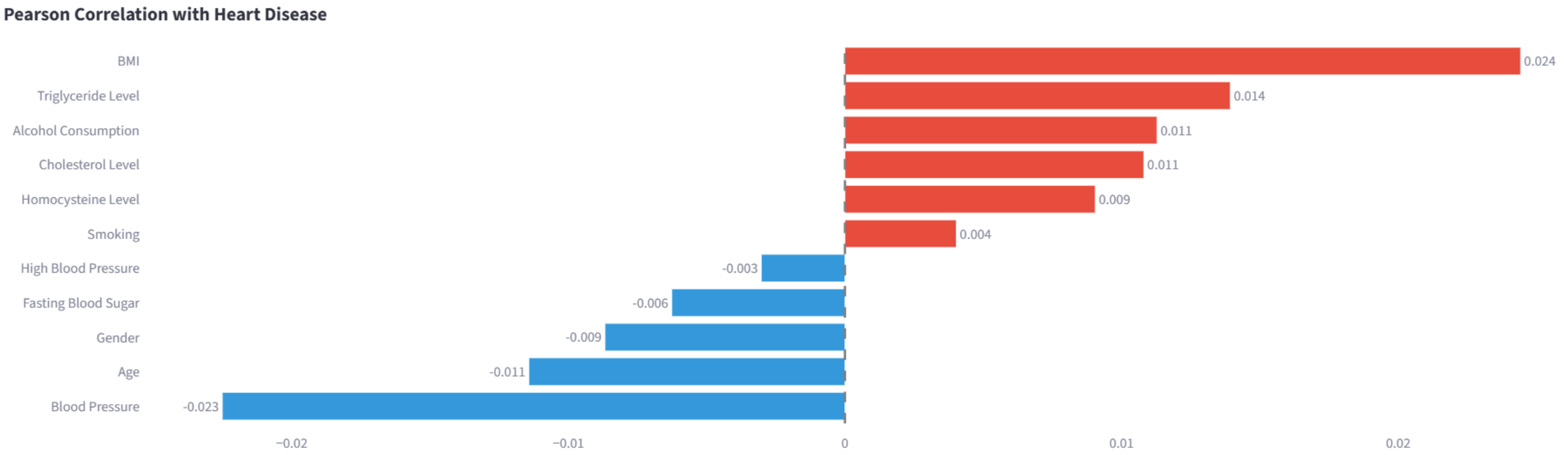


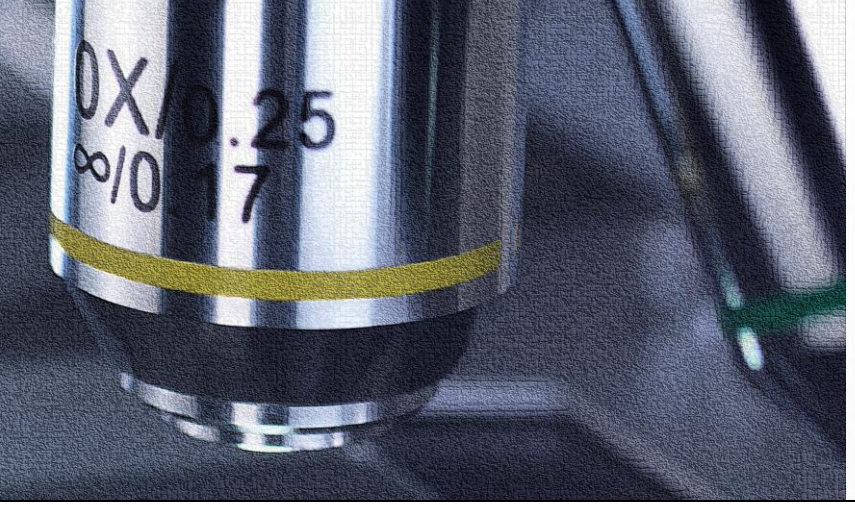


EDA

In the EDA phase, my key finding was that Pearson correlation coefficients were near zero (under 0.05). This was initially frustrating, but I realized that medical indicators often have non-linear relationships. For example, both high and low blood pressure pose risks. Simple linear correlation masks true medical patterns. Furthermore, I applied Feature Engineering. Based on medical thresholds, I converted continuous variables (like Age, BMI, and Triglycerides) into ordinal variables (levels 1 to 5, e.g., Youth to Elderly, Normal to High). However, this transformation did not yield improvements in model performance. This prompted me to shift to a machine learning-based visualization approach.

1. Pearson Correlation (Linear Association)





EDA

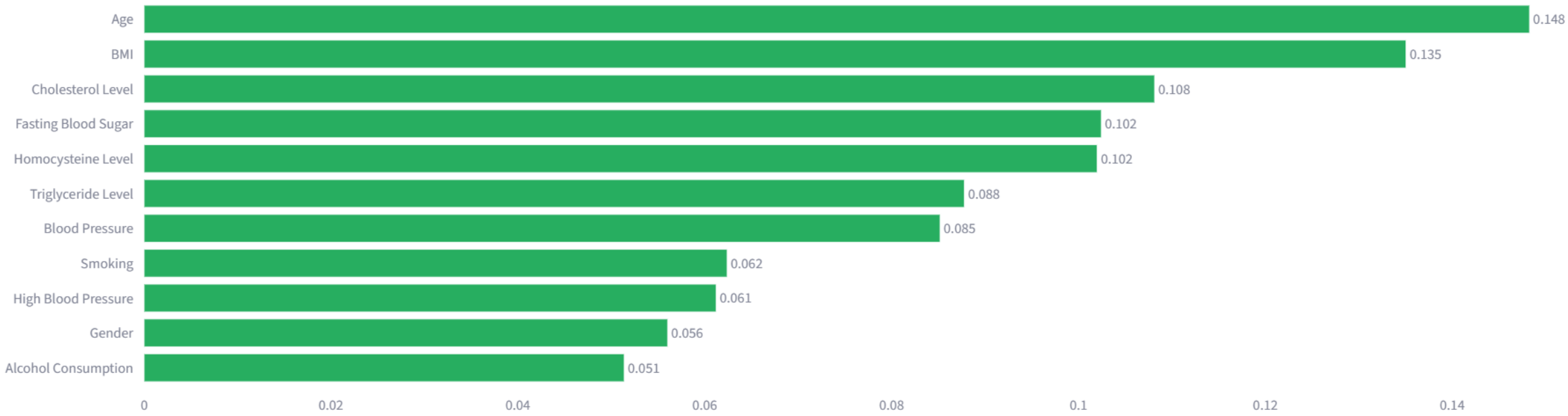
Random forest is selected for feature importance analysis, as it can capture nonlinear relationships and interactions.

Age is the dominant predictor, followed by BMI, Cholesterol, and Homocysteine. This perfectly match with medical consensus.

🌲 2. Random Forest Feature Importance (Non-Linear)

Why Random Forest Feature Importance? Pearson correlation only captures linear relationships. In medicine, risk factors often have complex, non-linear interactions. **Feature Importance** from a Random Forest model measures how much each variable contributes to the model's predictive accuracy. *(Note: The model is trained with* `class_weight='balanced'` *to account for class imbalance).*

Random Forest Feature Importance





RESULT

Main conclusion

1. Age is the dominant predictor for heart disease.
2. Weak linear correlation does not mean variables are useless. Non-linear models revealed underlying patterns.

Limitations

1. This is cross-sectional data; correlation is not causation.
2 .This is NOT a medical diagnostic tool and cannot replace clinical evaluation by a doctor.

Thank you